

Online Appendix

Educational Ambition, Marital Sorting, and Inequality

Frederik Almar, Benjamin Friedrich, Ana Reynoso, Bastian Schulz and Rune Vejlin

A Additional Details on Data and Measurement

A.1 Data sources

All registers used are yearly population-wide data sets. We use all the persons living in Denmark at the end of a year from 1980-2018, who are observed in the data sets PERSONER and BEF. These data sets contain yearly information on age, partner ID, municipality, gender, civil status, and number of children. We merge these data sets with additional information as follows. We measure the highest achieved education from the education register (UDDA). From the income register (IND), we measure income and wealth information, and the hourly wage earned in the primary job held in the last week of November each year (from IDAN).¹⁷ The datasets EXPYEAR and IDAP provide information on real labor market experience. Finally, we use information from the registers RAS and AKM in order to get occupational information and a part-time/full-time indicator. When available we also merge information from the Labor Force Survey in order to get more details on flexibility and hours worked than what is available from the registers. The Labor Force Survey covers the years 2000 to 2018. We keep information on individuals aged 19 to 60.

A.2 Definition of key variables

Main Income measure Our main income variable, ERHVERVSINDK_13, measures all earned income during a year, where earned income is defined as income from employment and self-employment.

In an extension, we consider annual disposable income, which is defined as the sum of total labor income (the income variable in the main analysis), total capital income, government transfers, and the predicted rental value of self-owned housing (had it been on the rental market) minus taxes, expenditures on interest rates, and alimony for children or spouses.

¹⁷We rank job types and keep the highest rank available in JOB_TYPE: H, 3, A, S, M. The variable for the hourly wage is TIMELON prior to 2008 and SMAL_TIMELOEN in 2008 and after.

Hourly wage Hourly wages are imputed from administrative information on labor income and hours worked in the employment register. Hours worked reflect contractual hours, i.e., part-time work is captured but not overtime. Before 2008, the hours are reported in four discrete bins. For further details, see [Lund and Vejlin \(2016\)](#). After 2008 we have the exact number of contractual hours.

We run a regression of log hourly wages on year dummies and educational specific experience profiles in order to take into account an aging population and differences in educations over time. We then subtract the coefficients on the year dummies (2000 base) from the log hourly wages. This gives us our residualized log hourly wages, which is what we use for the analysis.

Education We find the highest completed education of the individuals when they are the oldest (if not reaching 35 in the data) or when they are age 35. This is the program and year of graduation we use as their (final) educational program.¹⁸ We do not consider programs that an individual did not complete. For example, the most advanced program of high school dropouts is “Primary Education.”

Educational programs As a point of departure each educational program is an ISCED code. However, in a few places we change the definition slightly. In the start of the sample we have a group of individuals who have only 7th or 8th grade, because compulsory schooling ended in 7th grade until around 1960. We pool 7th-9th grade into one group called 9th grade. We split up both 9th grade (1109) and 10th grade (1110) into 5 sub programs each based on region of graduation.¹⁹ Finally, some individuals have an older code for high school than those graduating in 1980 and later. We assume that the high school education did not change much and we use starting wages and wage growth for the new high school education for those who graduated with the old code prior to 1980.

Starting wages and growth Starting wages are the average of the log hourly residualized wages in years 1-5 after graduation. The growth is calculated based on the difference between the average in years 9-11 and the starting wages. This is done for each individual in the sample (both singles and couples). In order to get information on the program level we average across individuals, but condition on individuals who graduated in 1980 or later, for whom we observe both for starting wages and having wages in some of the years 9-11 after graduation. We also

¹⁸Note that ambition types are assigned based on the *final* program, e.g., time-invariant, whereas educational level and field types are assigned based on the highest degree achieved by a given age. This is to be consistent with the previous literature on education-based marriage market types. In practice, this distinction has a neglectable impact because most individuals finish their studies in their 20s.

¹⁹In particular we do the following. Use 9th grade (1109)(split by region) starting wages and growth for the following codes (all also split by region depending on where the individual lives in the first year we see them in the data, e.g., 1980): 1007,1008,1023,1123,1009,1022. Use 10th grade (1110) (split by region) for: 1010.

only use information from individuals, whose wage growth is below the 99th percentile (extreme values are likely due to measurement error).

With this in place, we can standardize starting wage and growth. All individuals in the data have been assigned the average values from their final program. We generate the standardized variables by subtracting the mean and dividing by the standard deviation.

Next, we construct our four ambition types by using k-means clustering on the standardized starting wages and growth. All individuals are still in the data set, but because everybody from the same program has the same value, we are grouping at the program level.

Work hours and flexibility We use survey responses about usual working hours from the Labor Force Survey (LFS) to construct measures of typical work schedules by program. The LFS starts in 2000 and we observe at least 50 survey responses of prime-age workers (aged 25-54) for 724 unique educational program identifiers.

Using these data, we first measure the share of graduates working *short hours*, defined as the fraction of prime-age graduates by program who work part-time. Second, we construct the share working *irregular hours*, defined as the fraction of prime-age graduates by program who report evening, weekend, or overtime work.

We then merge these program-specific measures of work schedules and hours flexibility onto the population register 2000-2018, capturing about 95% of the baseline population sample. We standardize the shares of short and irregular hours and then use our k-means clustering methodology to group the 724 programs based on their similarity in these two hours-based dimensions. Assuming stable program characteristics on work schedules over time, we assign the resulting type classification by program to individuals over the earlier part of the sample, 1980–1999 as well.

Lifetime earnings We construct average lifetime earnings based on program graduates who are observed in the labor market for at least 30 years from the time of graduation. To compute lifetime earnings, we deflate earnings by running a regression of log annual earnings, for each program separately, on year dummies (base year 2000) and dummies for years since graduation to account for compositional differences by programs in the share of graduates at different life-cycle stages.

We can measure average lifetime earnings for 502 programs that have at least 30 workers observed in the register data over 30 years since graduation. We standardize the average lifetime earnings measure and then use our k-means clustering methodology for this single dimension to group programs by their similarity in average lifetime earnings.

A.3 Descriptive statistics

Table A.1: Basic Descriptive Statistics for Educational Ambition Types

Category (w_0, g)	(low, low)	(high, low)	(low, high)	(high, high)	Population
Population share	20.2%	22.7%	47.5%	9.7%	100.0%
Female share	64.8%	31.0%	56.0%	33.4%	50.0%
Starting wage	4.841 (0.0613)	5.015 (0.0775)	4.728 (0.0488)	5.181 (0.134)	4.860 (0.170)
Wage growth	0.0807 (0.0339)	0.118 (0.0436)	0.211 (0.0574)	0.301 (0.0756)	0.172 (0.0862)
Parental wealth at graduation	401347.0 (259668.7)	664844.4 (1609532.9)	269760.8 (307755.7)	1189937.8 (353775.9)	474762.7 (858804.7)
Wage growth SD	0.323 (0.0682)	0.298 (0.0536)	0.430 (0.0946)	0.365 (0.0731)	0.359 (0.0945)

Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The four first columns report averages of individual-level descriptive statistics for each of the four educational ambition types identified in Section 3. The final column reports the same statistics for the entire population of couples as defined in Section 2. Starting wages are measured in logs and wage growth are growth rates in hourly wages in the first ten years after graduation. Parental wealth at graduation is computed as the sum of both parents' net wealth in the year in which the individual graduates from the most advanced educational program. Deflated with base year 2000. Standard deviations in parentheses.

Table A.2: Educational Levels, Fields, and Ambition Types

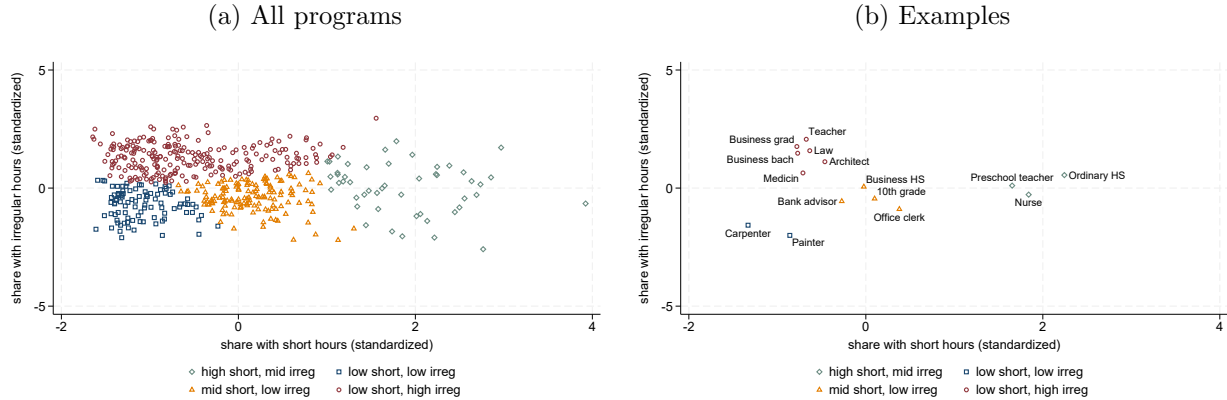
Category (w_0, g)	(low, low)	(high, low)	(low, high)	(high, high)	Population
<i>Educational Level</i>					
Primary	8.3%	0.5%	56.2%	0.2%	28.5%
Secondary	66.2%	57.3%	40.1%	10.3%	46.4%
Tertiary	24.9%	42.1%	8.2%	89.3%	24.9%
<i>Educational Level within Tertiary</i>					
Bachelor	24.1%	29.4%	3.1%	30.3%	15.9%
Master & PhD	0.8%	12.7%	0.5%	59.0%	9.0%
<i>Educational Field within Tertiary</i>					
Humanities	2.2%	18.0%	1.2%	2.7%	5.4%
Social Science	0.1%	3.0%	0.5%	16.4%	2.5%
Business	0.3%	0.5%	0.3%	21.4%	2.4%
STEM	0.2%	3.9%	0.2%	34.3%	4.4%
Health & Welfare	18.5%	12.3%	1.1%	11.6%	8.2%
Other	3.7%	4.4%	0.3%	3.0%	2.2%

Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The four first columns and the first panel report population shares for each of the four educational ambition types identified in Section 3 across educational levels. We further subdivide the tertiary shares into Bachelor and Master/PhD as well as post-secondary fields of study. The final column reports the all shares for the entire population of couples as defined in Section 2.

A.4 Clustering on Work Hours or Lifetime Income

This section presents the results for our alternative clustering approaches based on hours flexibility or lifetime earnings and compares them to the baseline ambition type classification.

Figure A.1: Educational Hours Flexibility types and their clustering variables

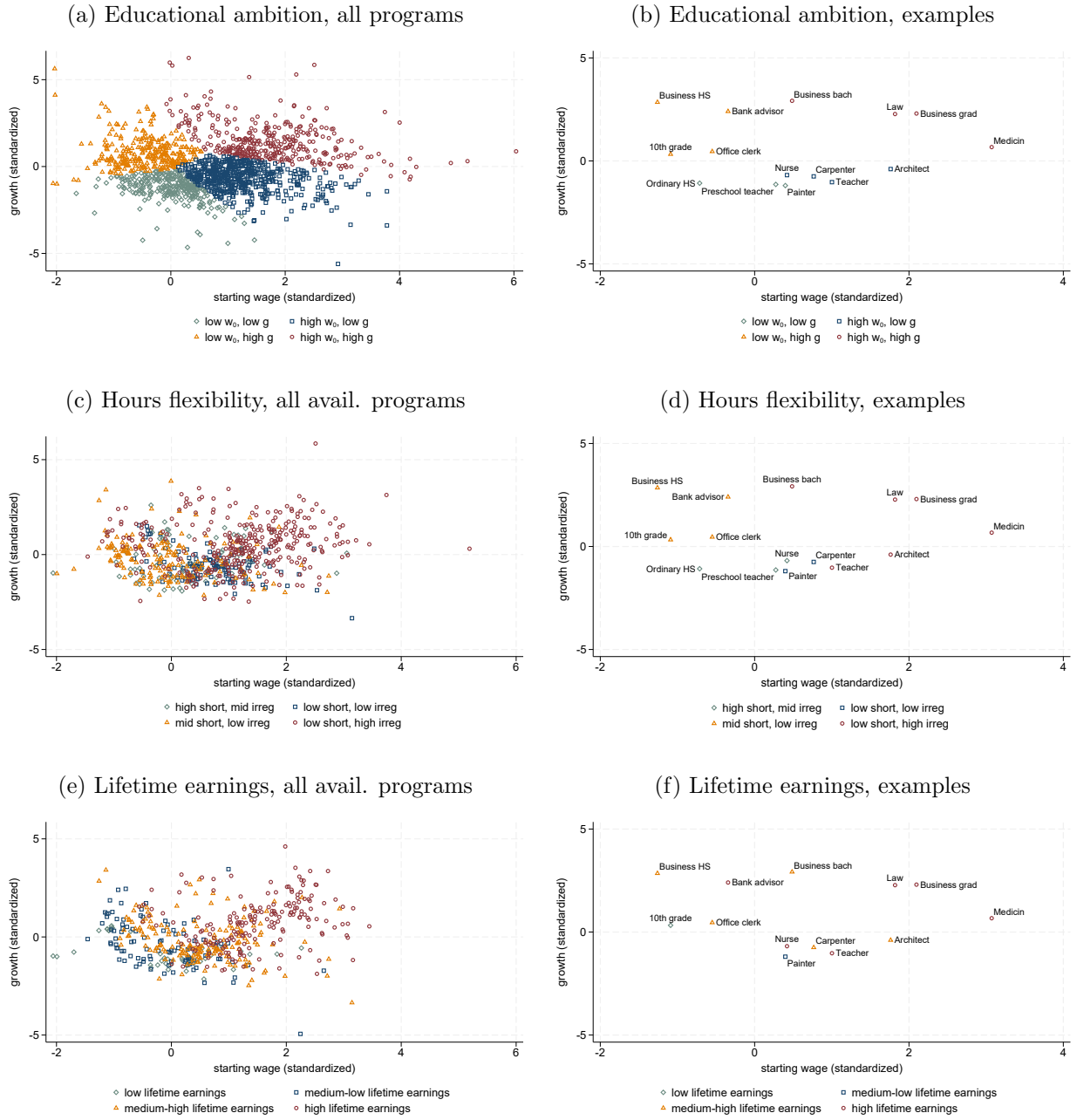


We start by presenting the results for clustering on the share of program graduates who work part-time and whose work schedule includes irregular hours. Panel (a) of Figure A.1 shows the resulting four clusters, while Panel (b) provides examples of programs in each cluster. The red-circle cluster consists of programs whose graduates typically have a low frequency of part-time work and a high frequency of irregular hours. Popular programs in this group include Business, Medicine, and Law. In contrast, the gray diamond cluster consists of programs whose graduates show a high frequency of short hours and have a lower prevalence of irregular hours. Examples include preschool teachers and nurses.

The classification based on lifetime earnings uses a one-dimensional k-means clustering approach and groups programs into low, medium-low, medium-high, and high income groups.

Comparison to Ambition Types In order to illustrate how the classifications based on lifetime earnings and hours flexibility relate to our baseline ambition-type classification, we compare all categorizations in the (w_0, g) plane in Figure A.2. Panel (a) shows the ambition-type clustering and can be compared to Panels (c) and (e) for hours-based and income-based types, respectively. In addition, Panels (b), (d), and (f) provide the types for the largest educational programs for each classification, respectively. For hours flexibility, we find substantial differences in particular for programs with low wage growth. Notably, the set of “inflexible” programs with low part-time share and high hours irregularity (low short, high irregular) also includes many programs with low wage growth, and not only programs with high wage level and growth (high w_0 , high g). For example, the hours clustering groups teachers and architects in the same “inflexible” cluster that also contains lawyers and doctors. In contrast, the wage-

Figure A.2: Baseline vs. Hours-Flexibility and Lifetime-Earnings Types



Notes: The horizontal axis is w_0 (standardized average starting wage) and the vertical axis is g (standardized average wage growth)

based categorization treats teachers differently from lawyers because of their different wage dynamics, with teachers being similar to nurses and carpenters. Yet, the hours classification assigns these three programs into three different groups because of their differences in work schedules: nurses with a high part-time share, teachers with higher reported workloads at the evenings and weekends, and carpenters with the highest full-time share but regular schedules. These differences will turn out to be important in explaining differences in assortative matching and counterfactual inequality trends, as we discuss below.

For lifetime earnings (LE), the comparison shows that in particular low-low ambition types are spread across many different income groups. For this reason, the LE categorization cannot pick up increased assortative matching among low types. For high ambition types, we find more overlap with LE types but we note that some programs with substantially different wage profiles (e.g., nurse, teacher, business grad, lawyer) fall into the same, highest earnings group (compare Figure A.2 Panels (b) and (f)).

A.5 Wage dynamics predict career and family outcomes

To analyze these career patterns further, we use regression analysis at the program level and show in Table A.3 that the two building blocks of our ambition types—the wage measures w_0 and g capturing returns to human capital in the labor market —jointly explain key career and family outcomes, even when comparing graduates within the same education level or field of study. We further control for the measures of hours flexibility and life-time earnings, to probe the extent to which the wage measures capture the role of work schedules or total earnings, or reflect labor market returns whose influence on career and family outcomes cannot be fully captured by work hours or lifetime earnings.

To show this, we build on the literature and construct seven proxies for career and family outcomes of graduates of specific educational programs. The proxies emphasize the trade-off between career investments and time commitments to the family (Wiswall and Zafar, 2021; Goldin, 2014; Calvo et al., 2024).

Part-time penalty is constructed as the ratio of the average wage w_1 of full-time workers to that of part-time workers.²⁰ It reflects the additional return to working long hours, a measure of inflexibility inspired by Claudia Goldin’s analysis of within-occupation gender differences in the US (Goldin, 2014). *Ever manager* is the fraction of graduates who reach a managerial position (hold a corresponding occupational code for at least two consecutive years). *Participation* captures the average share of time across the life-cycle during which program graduates are active in the labor market (work at least part-time). *Full-time* captures the fraction working at least 32 hours per week.²¹ *Age at first child* is the average age among graduates at which the first child is born. *Wealth at age 50* is the average net wealth accumulated at age 50 in Danish Crowns (henceforth DKK).²² Finally, *Lifetime earnings* is the sum of deflated annual earnings over 30 years after graduation.

²⁰Recall that w_1 was the average wage in years 9-11 in the labor force used to construct the ambition types.

²¹Based on the RAS register. Before 2008 the threshold between part-time and full-time is at 30 hours (1980 to 1992) or 27 hours (1993 to 2007), see Lund and Vejlin (2016).

²²Net wealth excludes assets in pension funds and is deflated with 2000 as the base year.

Table A.3: The Work-Life Balance of Ambition beyond Levels, Fields, and Lifetime Earnings

FE model:	Levels	Fields	Levels		Levels	Fields	Levels	
<i>Controls:</i>	<i>None</i>	<i>None</i>	<i>Earnings</i>	<i>Hours</i>	<i>None</i>	<i>None</i>	<i>Earnings</i>	<i>Hours</i>
	(a) Part-time penalty				(b) Ever manager			
w_0	0.0072 (0.007)	0.0030 (0.005)	0.0294 (0.006)	-0.0105 (0.010)	0.0241 (0.005)	0.0127 (0.005)	0.0227 (0.007)	0.0144 (0.006)
g	0.0208 (0.005)	0.0160 (0.005)	0.0323 (0.005)	0.0136 (0.006)	0.0262 (0.002)	0.0190 (0.002)	0.0294 (0.003)	0.0214 (0.003)
Mean	1.098				0.051			
Obs.	985	985	443	684	1,874	1,874	496	697
Adj. R^2	0.316	0.480	0.450	0.373	0.409	0.467	0.450	0.529
	(c) Participation				(d) Full time work			
w_0	0.0377 (0.009)	0.0278 (0.014)	-0.0143 (0.008)	0.0288 (0.013)	0.0955 (0.013)	0.0844 (0.016)	0.0703 (0.009)	0.0201 (0.011)
g	0.0366 (0.007)	0.0352 (0.009)	0.0151 (0.007)	0.0342 (0.008)	0.0212 (0.008)	0.0194 (0.011)	0.0131 (0.006)	-0.003 (0.005)
Mean	0.764				0.821			
Obs.	1,874	1,874	496	697	1,874	1,874	496	697
Adj. R^2	0.449	0.466	0.664	0.471	0.314	0.293	0.494	0.560
	(e) Age at first child				(f) Wealth at age 50			
w_0	0.529 (0.366)	0.561 (0.366)	0.153 (0.460)	0.414 (0.455)	0.144 (0.016)	0.134 (0.016)	0.135 (0.016)	0.105 (0.017)
g	0.277 (0.185)	0.339 (0.218)	0.0269 (0.190)	0.154 (0.203)	0.0945 (0.013)	0.0864 (0.015)	0.0858 (0.013)	0.0777 (0.014)
Mean	31.61				0.243M			
Obs.	1,824	1,824	493	692	1,305	1,305	496	647
Adj. R^2	0.012	0.015	0.143	0.021	0.465	0.472	0.539	0.479

Notes: *FE* stands for *fixed effects*, *Obs.* for *observations*, and *Adj.* for *adjusted*. w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. *Earnings* stands for *Life-time earnings* as defined in the text of this section. Each panel (a) to (f) shows the coefficients on w_0 and g in a regression of the work-life balance proxy (defined in the text of this section), which includes the fixed effects and controls as indicated in the columns' labels. Robust standard errors in parentheses.

Table A.3 shows the coefficients from regressions of the first six proxies (we control for life-time earnings in the fourth column of each Panel) on w_0 and g . The first column of each Panel includes educational-level fixed effects to compare programs within the same level. The second column instead uses fixed effects for field of education. The third and fourth column respectively combine level fixed effects with controls for lifetime earnings (col 3) or controls for the share of graduates working short hours and the share working irregular hours (col 4), as defined in Appendix A.2.

For all proxies, we find that higher starting wages or higher wage growth (and in most cases both) are associated with more career focus, which affects the work-life balance negatively. This is true even within the same levels or fields of study. For example, a one standard deviation change of wage growth (recall that w_0 and g are standardized) implies that the inflexibility

measure (part-time penalty) increases by 2.1% *relative* to the mean across programs. Within the same level or field of education, the effects become somewhat smaller (1.9% and 1.5%, respectively) but remain highly significant.

Similarly, we find that higher starting wages and wage growth are associated with higher labor force participation and full-time work, as well as higher chances of manager promotion and higher net wealth, but also higher age at first child. Higher wage growth is associated with higher degrees of inflexibility (part-time penalty) across all specifications, while starting wages are less so. This is expected as wage growth is most likely associated with higher degrees of investment in human capital. Finally, age at first child is strongly associated with starting wages and wage growth in all specifications except when controlling for life-time earnings where the association with wage growth disappears. All these patterns emphasize important heterogeneity in labor market and family outcomes across programs that our ambition types can capture within educational levels and fields.

Moreover, the third specification in each model shows that starting wages and wage growth correlate with our measures of work-life balance even conditional on discounted lifetime earnings. The main takeaway from this result is that collapsing the multi-dimensional labor market trajectories of individuals into a one-dimensional measure of earnings misses the interplay between labor market starting conditions and growth trajectories. Different combinations of starting wages and wage growth can have opposite implications for work-life balance. Consider that the same level of lifetime earnings can be reached through a high starting wage and relatively low growth, or a low starting wage and relatively high growth. Indistinguishable by means of lifetime outcomes, these different combinations imply, e.g., different working hours. Thus, they have different values from the perspective of marriage and family, which is captured by different ambition types. Capturing this heterogeneity in the family value of different career types is a key advantage of our new categorization.

Finally, we find in the fourth column of each panel that ambition types strongly predict career outcomes and correlate with work-life balance outcomes in part through labor-supply patterns. Compared to the previous regression models, the coefficients on our wage moments decline somewhat in absolute value for several of the career and family outcomes when controlling for the share of workers with short or irregular hours. While the wage moments typically remain statistically significant, the hours moments are also significantly related to the family and career outcome measures. As such, the results confirm the notion that labor market returns imply opportunity costs for the family and lead to different labor supply choices.

A.6 Weights for the Aggregate Measure of Assortative Matching

Almar and Schulz (2024) derive a decision rule to minimize the distortion of likelihood-ratio-based measures due to changing marginal distributions. While accounting for changing population shares ensures proper scaling of the likelihood ratios (see equation (1)), the marginal distributions also influence the measure beyond this desired feature. To minimize this distortion, Almar and Schulz (2024) propose to set λ in equation (3) such that the combined distortion from changing male and female marginal distributions is minimized. In our data, the educational attainment of females was initially lower but increased relatively more compared to males. Thus, to minimize the distortion, it is optimal to put the weight on the “short side” of the market (high-type females). Recall that λ ($1 - \lambda$) is the parameter of the male (female) marginal in the convex combination, and let γ_1 (γ_2) be the total distortional effect for men (women) in the total differential of the aggregate measure (2). The optimal λ is

$$\lambda^* = \begin{cases} 0 & \text{if } \text{sign}(\gamma_1) = \text{sign}(\gamma_2), \quad |\gamma_2| > |\gamma_1| \\ \frac{\gamma_1}{\gamma_1 - \gamma_2} & \text{if } \text{sign}(\gamma_1) \neq \text{sign}(\gamma_2) \\ 1 & \text{if } \text{sign}(\gamma_1) = \text{sign}(\gamma_2), \quad |\gamma_1| > |\gamma_2|. \end{cases} \quad (\text{A.1})$$

Intuitively, if the signs of the total distortional effect are similar for men and women (e.g., the share of “high-type” individuals could be increasing for both genders), the weight is set such that the bigger change (in absolute terms) is taken into account by the aggregate measure. For example, if $\gamma_2 > \gamma_1$, female type distribution changes are relatively more important. To reflect this, changes of the male marginal distribution are taken out by setting λ equal to 0. The female marginal distribution cancels out in the likelihood ratios (1), removing its distortional impact. For the intermediate case $\text{sign}(\gamma_1) \neq \text{sign}(\gamma_2)$, $\lambda \in (0, 1)$ ensures that both population share changes are reflected. The optimal λ^* may change over time, depending on the changing configuration of the marginals. Thus, the optimal weights may flip when, e.g., the population share of high-type women becomes greater than the population share of high-type men. To ensure that trends in assortative matching are comparable over time, we set λ to the modal value suggested by (A.1) over time. In the vast majority of years and for all categorizations, the rule suggests putting the weights on female type distribution changes.

A.7 Counterfactual Scenarios and Assortative Matching

In this appendix section, we provide the formal derivation of the result that our choice of weights in Section 5.1, scenario (i), fixed household composition, guarantees that those who are married in year τ exhibit the same degree of assortative matching S as the 1980 households.

In addition, the log odds ratios are also held fixed at their 1980 values.

To see this result, first note that our choice of weights for $\tau_y = \tau$ and $\tau_\mu = 1980$ is

$$\hat{\psi}_{\tau,80}(i,j) = \frac{\frac{N_{80}(i,j)}{N_{80}^{\text{all}}}}{\frac{N_\tau(i,j)}{N_\tau^{\text{all}}}} = \frac{N_{80}(i,j)}{N_\tau(i,j)} \times \frac{N_\tau^{\text{all}}}{N_{80}^{\text{all}}} = \frac{P_{80}^M(i,j)}{P_\tau^M(i,j)} \times \underbrace{\frac{\sum_{r \neq \phi} \sum_{s \neq \phi} N_{80}(r,s)}{\sum_{r \neq \phi} \sum_{s \neq \phi} N_\tau(r,s)}}_{:=A} \times \frac{N_\tau^{\text{all}}}{N_{80}^{\text{all}}},$$

where $N_y(i,j)$ denotes the number of $(t_m = i, t_f = j)$ households (including singles if one of the types is $\emptyset$) and N_y^{all} is the total number of households in year y , while $P_y^M(i,j)$ denotes the fraction of married couples of type $(t_m = i, t_f = j)$ in year y .

Using equations (1) and (3) to write the aggregate measure S in equation (2) explicitly, and applying the weights $\hat{\psi}_{\tau,80}(i,j)$, yields the proposed result that $\tilde{S}_\tau = S_{80}$:

$$\begin{aligned} \tilde{S}_\tau &= \sum_{t=1}^T \frac{P_\tau^M(t,t) \times \hat{\psi}_{\tau,80}(t,t)}{\left(\sum_{s=1}^T P_\tau^M(t,s) \times \hat{\psi}_{\tau,80}(t,s) \right) \left(\sum_{r=1}^T P_\tau^M(r,t) \times \hat{\psi}_{\tau,80}(r,t) \right)} \\ &\quad \times \left(\lambda \sum_{s=1}^T P_\tau^M(t,s) \times \hat{\psi}_{\tau,80}(t,s) + (1-\lambda) \sum_{r=1}^T P_\tau^M(r,t) \times \hat{\psi}_{\tau,80}(r,t) \right) \\ &= \sum_{t=1}^T \frac{P_\tau^M(t,t) \times \frac{P_{80}^M(t,t)}{P_\tau^M(t,t)} \times A}{\left(\sum_{s=1}^T P_\tau^M(t,s) \times \frac{P_{80}^M(t,s)}{P_\tau^M(t,s)} \times A \right) \left(\sum_{r=1}^T P_\tau^M(r,t) \times \frac{P_{80}^M(r,t)}{P_\tau^M(r,t)} \times A \right)} \\ &\quad \times \left(\lambda \left(\sum_{s=1}^T P_\tau^M(t,s) \times \frac{P_{80}^M(t,s)}{P_\tau^M(t,s)} \times A \right) + (1-\lambda) \left(\sum_{r=1}^T P_\tau^M(r,t) \times \frac{P_{80}^M(r,t)}{P_\tau^M(r,t)} \times A \right) \right) \\ &= S_{80} \times \frac{A \times A}{A^2} = S_{80} \end{aligned}$$

By a similar argument, we can show that the log odds ratios remain fixed. Consider types $i = \{t(1), \dots, t(T-1)\}$ and $j = \{t(2), \dots, t(T)\}$ for any categorization t . For any 2×2 submatrix of joint types in which $i \neq j$,

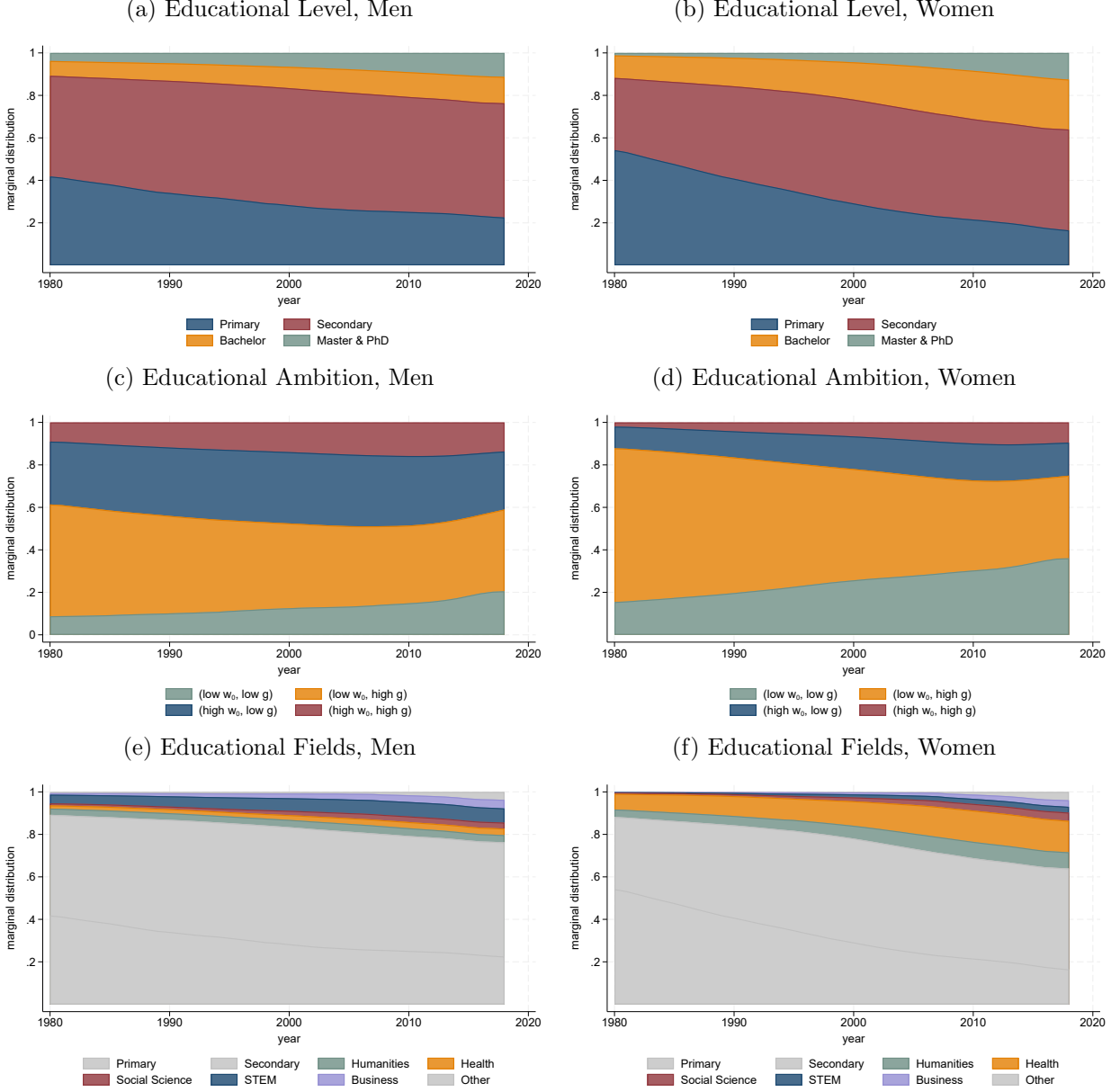
$$\begin{aligned} \tilde{I}_{\text{odds},\tau} &= \log \left(\frac{P_\tau^M(i,i) \times \hat{\psi}_{\tau,80}(i,i) \times P_\tau^M(j,j) \times \hat{\psi}_{\tau,80}(j,j)}{P_\tau^M(j,i) \times \hat{\psi}_{\tau,80}(j,i) \times P_\tau^M(i,j) \times \hat{\psi}_{\tau,80}(i,j)} \right) \\ &= \log \left(\frac{P_{80}^M(i,i) \times P_{80}^M(j,j)}{P_{80}^M(j,i) \times P_{80}^M(i,j)} \right) = I_{\text{odds},80} \end{aligned}$$

B Additional Results

B.1 Differences in Demographic Changes Across Categorizations

B.1.1 Trends in Education Distributions

Figure B.1: Marginal Type Distributions

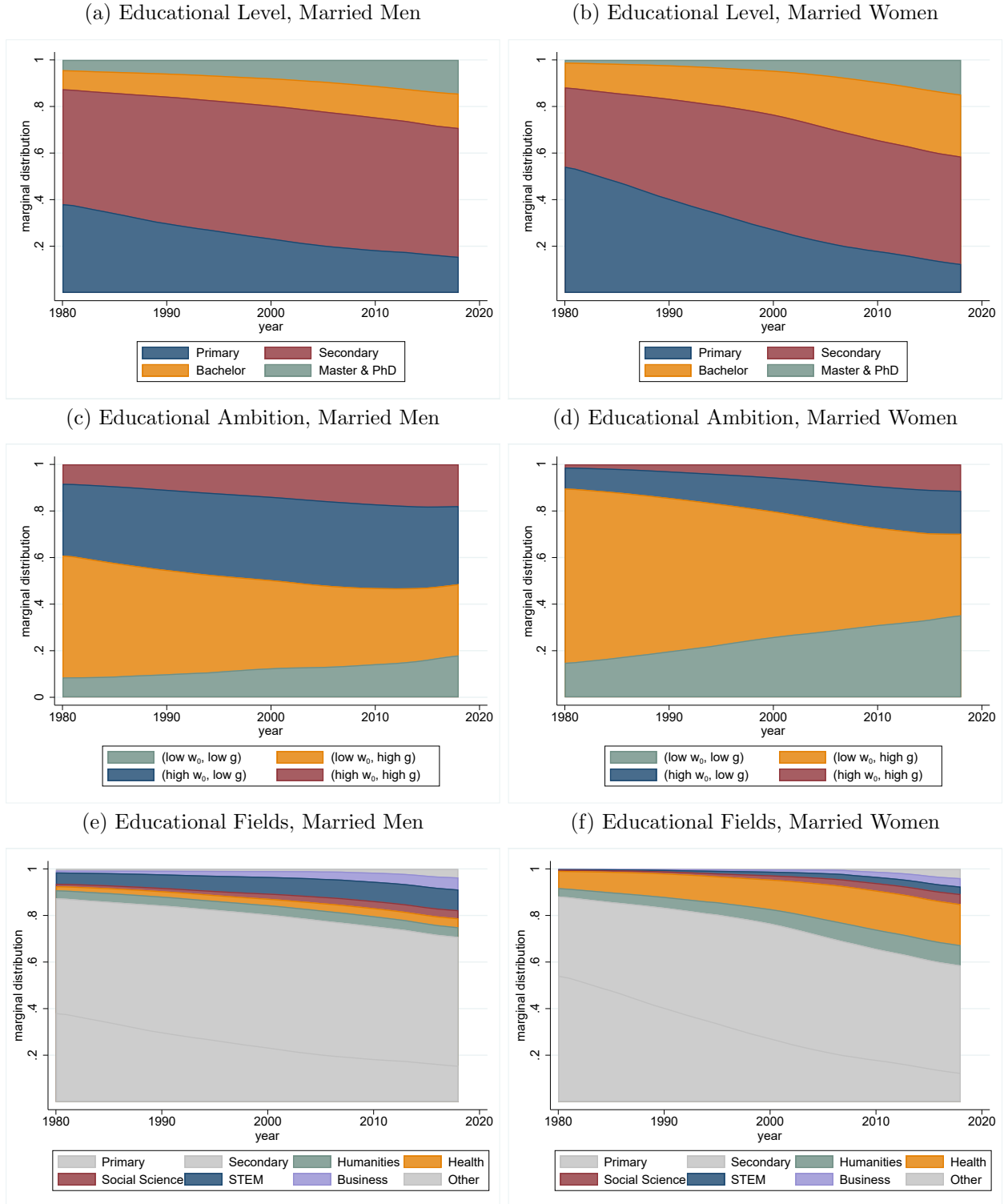


Notes: Marginal distributions for men and women over time by educational level and educational ambition. Sections 2 and 3 explain how the sample, educational levels, educational- fields and the labor market outcome (residualized log hourly wages) underlying the educational ambition types are constructed.

How we view educational attainment and its evolution differs depending on which categorization we consider. Figure B.1 shows the fraction of men (left panels) and women (right panels) by education type and year, for our three definitions of education—Level in panels (a) and (b), Ambition in (c) and (d), and Fields in (e) and (f). For educational levels, the “secondary” category is large and relatively stable for both men and women, while the most

frequent ambition types for both men and women are the ones with high wage growth. Notably, we observe increasing shares of individuals, especially women, in tertiary education relative to primary but increasing shares of individuals in both the bottom and top ambition types.

Figure B.2: Marginal Type Distributions, Conditional on Marriage

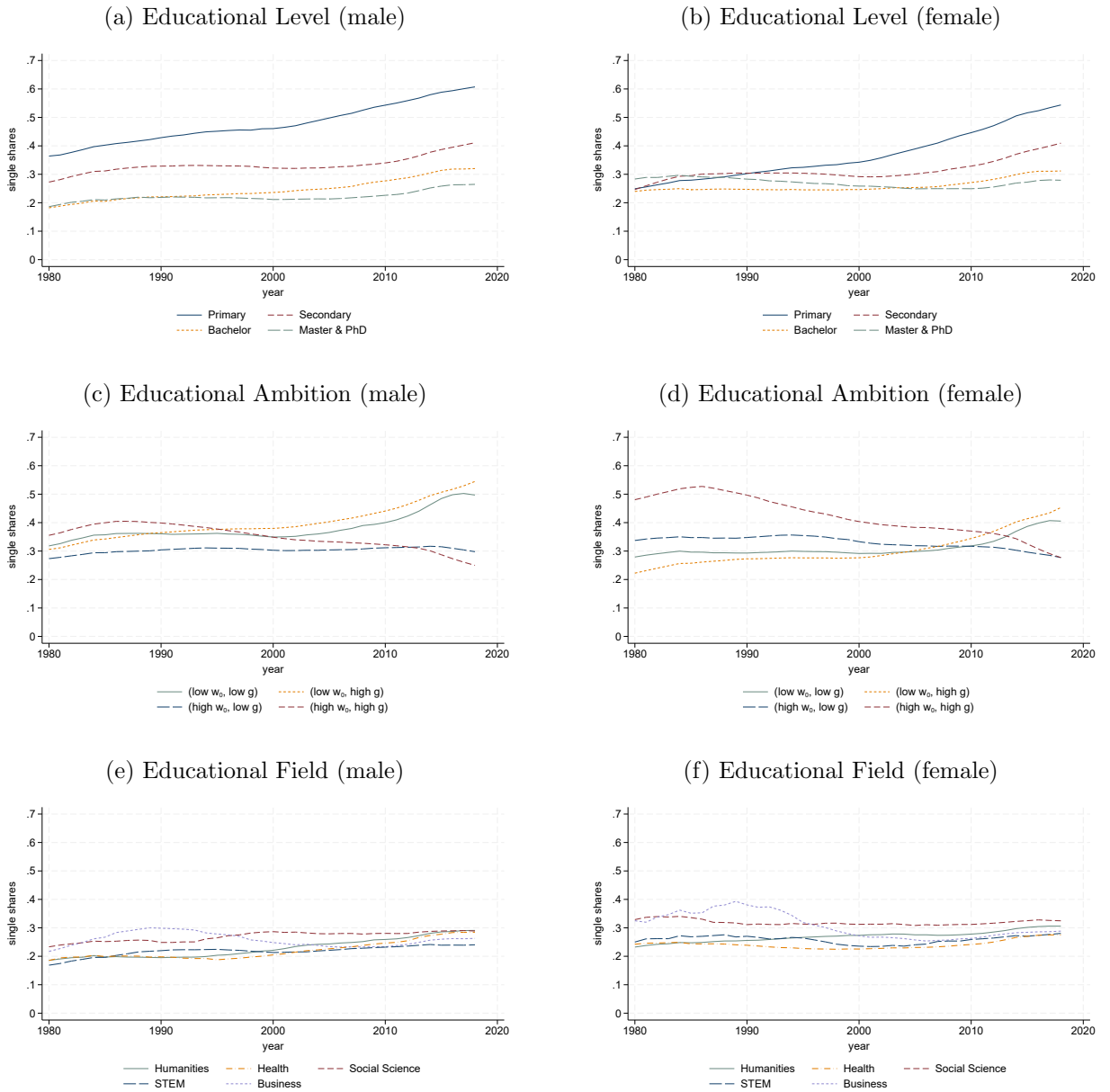


Notes: Marginal distributions for married and cohabiting men and women over time by educational level, educational ambition, and educational field. Sections 2 and 3 explain how the sample, educational levels, educational- fields and the labor market outcome (residualized log hourly wages) underlying the educational ambition types are constructed.

In addition, Figure B.2 shows the educational composition of married individuals over time. We observe that the composition of married individuals has shifted over time towards highly-educated (Master & PhD for both genders and also Bachelor degrees for women) and highly ambitious individuals (especially for women), but there are important differences across categorizations and gender. For example, based on ambition we also observe an increasing representation of low-ambition individuals among the married population.

B.1.2 Trends in Marriage

Figure B.3: Single Shares for Educational-Level, Ambition Types, and Fields of Study



Notes: Single shares by gender for all for types of the educational level (Panels a and b), educational ambition (Panels c and d), and educational field (Panels e and f) categorizations. Types are constructed as explained in Section 3.

We analyze how individuals sort into being married or single based on their education type and make comparisons across categorizations. In Figure B.3, we show how single shares of men and women have evolved for all the types included in our educational-level, educational-ambition, and educational-field categorizations.

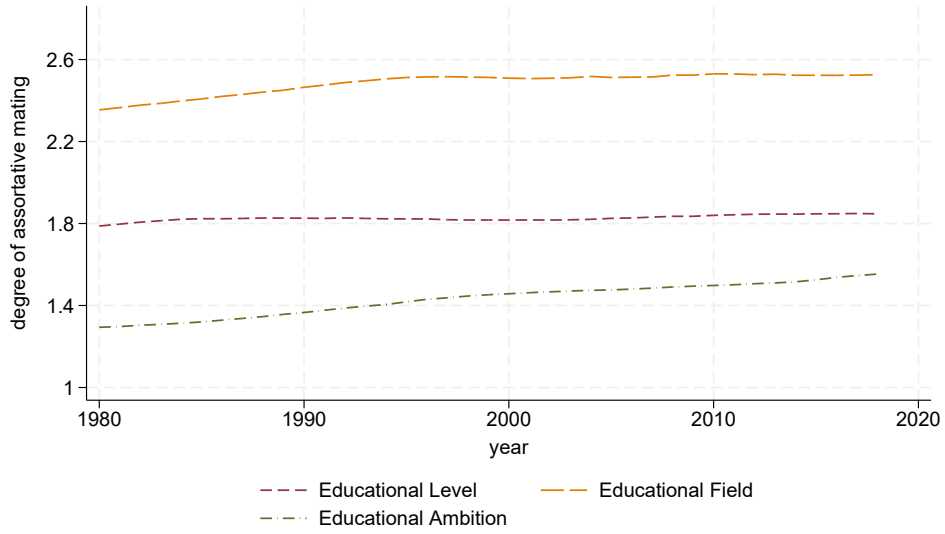
For educational-level types, single shares have increased for all types for both men and women, and most significantly so for individuals with primary education (blue solid lines in Panels (a) and (b), Figure B.3). In contrast, if we interpret the data through the lens of our novel ambition types, single shares did not increase for all types. Specifically, men and women who graduate from ambitious educational programs with high starting wages and high wage growth (red dashed lines) exhibit falling single shares. This is especially pronounced for high-type women, who had the highest probability of remaining single in 1980, when their single share was close to 50%. It decreased to around 30% in 2018. If more women with high earnings potential enter marriage, inequality between households might increase, depending on the partner type and couples' labor supply choices.

Single shares increased for both men and women who graduate from educational programs with low starting wages (gray solid and yellow short-dashed lines). The blue dashed lines—the single shares of men and women who hold degrees with high starting wage but low wage growth—changed relatively little but decreased slightly for women. These striking differences in single share trends across categorizations show that the ambition types reveal fundamental changes in marriage market matching that remain undetected using standard educational-level types and potentially matter for inequality trends.²³

²³The single-share trends by educational field are rather flat, see Panels (e) and (f). They are different from both educational levels and ambition, which underscores our point that the role that the extensive margin plays depends on the categorization.

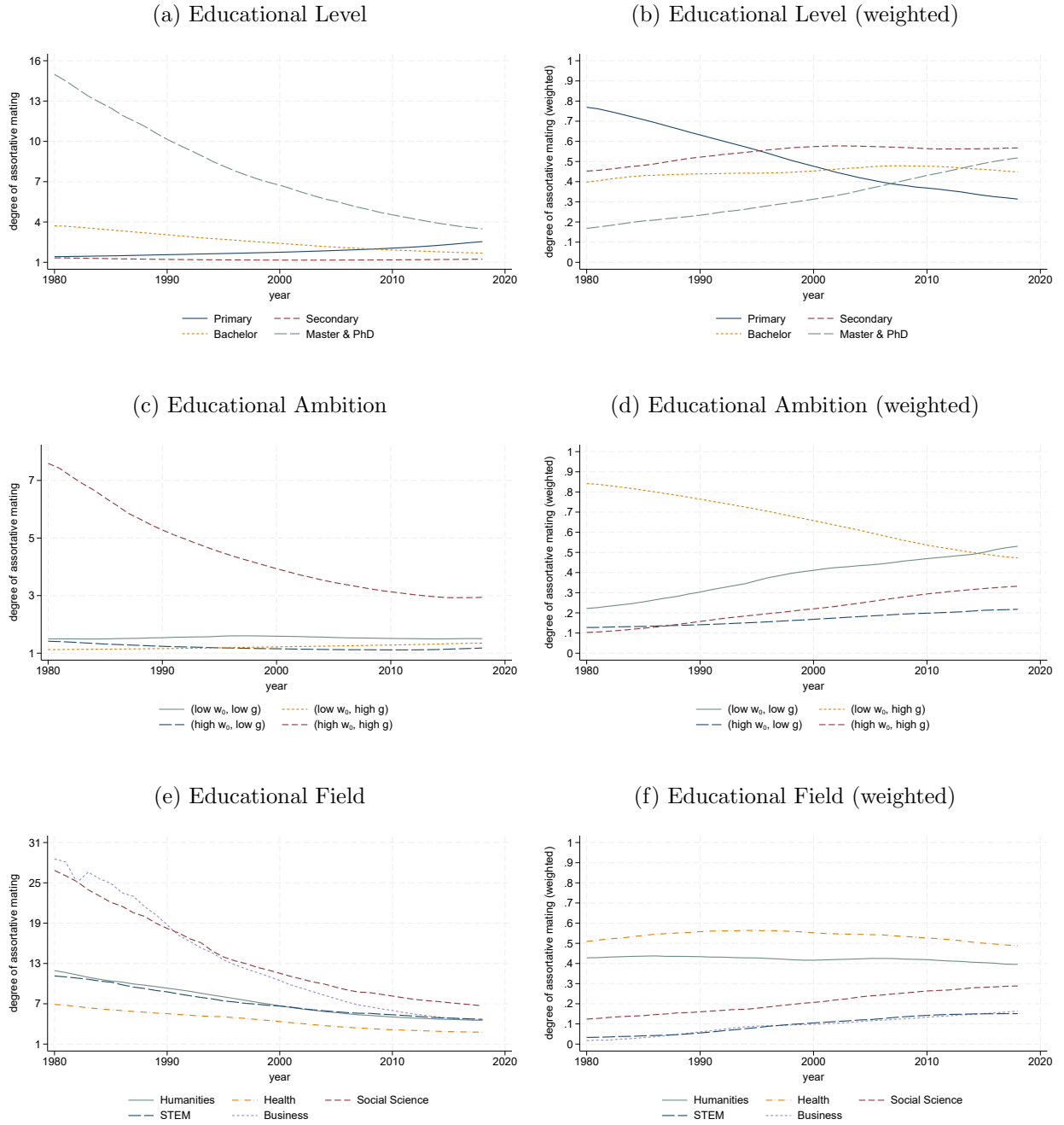
B.2 Trends in Measures of Assortative Matching

Figure B.4: Aggregate Measures in Levels



Notes: The figure shows the measure $\mathcal{S}$ derived in Section 4, equation (2) for educational level types (red short-dashed line), educational field types (orange long-dashed line) and educational ambition types (green dash-dotted line). Types are constructed as explained in Section 3.2.

Figure B.5: Type-specific Likelihood Indices, unweighted and weighted



Notes: Likelihood indices for equal-education couples (equation (1)) for educational level, educational ambition, and educational field categorizations. Types are constructed as explained in Section 3.2.

B.3 Work Hours and Lifetime Earnings

In this section, we compare our main results for assortative matching and inequality trends based on our ambition type classification to analogous results for the classifications based on work hours (abstracting from the returns to work hours and schedules) or lifetime earnings (abstracting from different career trajectories that lead to similar total income).

Yet, we note that both approaches are associated with significant measurement challenges. To construct hours types, we rely on survey responses from the LFS, which represent a small fraction of all workers in Denmark. We decide to include all programs with at least 50 survey responses in our analysis. The LFS is limited to 2000–2018, and we impute the hours type to the earlier part of the sample for each program, assuming stable hours characteristics.

To construct lifetime earnings, we need to observe program graduates for at least 30 years from the time of graduation. This leads to lower sample coverage of this alternative classification in early and late years of our analysis period, when a higher share of individuals have outdated or new degrees with missing information on lifetime earnings, respectively. This limitation affects couples more than singles because our analysis requires assigning types to both spouses to be included. As a result, we focus on a shorter time period from 1990 to 2010 for this robustness analysis because over this period the share of singles and couples covered by the lifetime-earnings classification is highest and stable. To interpret the magnitude of the results based on the lifetime-earnings classification, we also provide results for our benchmark ambition type classification for the 1990–2010 period. Specifically, we report results both for the smaller sample of households included in the lifetime-earnings analysis and for the full sample of households in these years.

B.3.1 Assortative Matching

Table B.1 present the results for hours in Panel (i) and lifetime earnings in Panel (ii), respectively. We start by analyzing the trends in assortative matching implied by each classification. For both alternatives, we find that the increase in PAM is much smaller than for our ambition types. Overall, this suggests that the increase we find is driven by returns to human capital. This cannot be detected by our lifetime earnings measure, and it is not consistent with an increase in the importance of work-life balance as captured by our hours flexibility measures.

First, while assortative matching on hours is positive (the aggregate measure is above one), the hours-based classification indicates only a small increase in PAM over time. As illustrated in Appendix A.4, there is a close relationship between wage dynamics and hours flexibility of

Table B.1: Assortative Matching and Fixed Household Composition for Hours-based and Lifetime-Earnings-based Classifications

	(a)		(b)			(c)		(d)	
	N (1,000s)		Assortative Matching			Gini, data		Gini, (i)	$\frac{\Delta_{Gini,(i)}}{\Delta_{Gini,data}}$
	1980	2018	1980	2018	Change	1980	2018	2018	
Ambition types	2,427	2,801	1.29	1.55	20.1%	0.298	0.400	0.364	64.6%
(i) Work hours and flexibility									
Hours flexibility	2,311	2,657	1.53	1.61	5.0%	0.298	0.401	0.367	66.8%
Ambition, hours sample	2,311	2,657	1.31	1.57	19.1%	0.298	0.401	0.363	63.0%
(ii) Lifetime earnings sample 1990 – 2010									
	1990	2010	1990	2010	Change	1990	2010	2010	
Lifetime earnings	1,555	1,578	1.52	1.56	3.0%	0.332	0.353	0.344	57.1%
Ambition, LE sample	1,555	1,578	1.40	1.55	11.0%	0.332	0.353	0.345	64.7%
Ambition, full sample	2,676	2,705	1.37	1.50	9.7%	0.327	0.363	0.347	55.4%

Notes: Columns (a) display the number of observations in each case in thousands of individuals for both 1980 and 2018. Columns (b) show our measure of assortative matching S derived in Section 4 for both years along with the percentage change. Columns (c) show the observed Gini coefficients in each case for 1980 and 2018. Columns (d) show the counterfactual Gini coefficient in the fixed marriage market scenario (i), i.e., had the composition of households stayed fixed at its 1980 level in each case and the fraction of the observed change in Gini that is captured by the change in the counterfactual scenario, $\Delta_{Gini,(i)}/\Delta_{Gini,data}$. Each row shows the columns' statistic for one of the alternative definitions of ambition types.

different educational programs. Yet, the hours-based classification focuses on the number of hours and their irregularity, not on their returns. This distinction is salient in Figure A.2 where the hours classification treat teachers and lawyers the same because of similar hours profiles, but they fall into different ambition types because of the different wage dynamics. These differences matter because the substantial increase in positive assortative matching based on ambition types is partly driven by programs with the highest career potential.

Second, Panel (ii) shows similar results for lifetime earnings, thus abstracting from the heterogeneity in starting wages and wage growth conditional on lifetime earnings: we find only a small increase in AM by about 3% over 1990–2010 based on lifetime-earnings types. This analysis includes about 58% of the sample in both 1990 and 2010, with similar coverage of singles and couples. In contrast, the increase implied by our benchmark types is much larger, at 11% in the same sample, or 9.7% in the full sample over this period, amounting to about half of the total increase in AM that we measure over four decades in the main results (20.1%).

B.3.2 Inequality

Turning to inequality trends in columns (c) and (d) of Table B.1, we find that both hours-based and lifetime-earnings types can capture a similarly large role of the household composition in inequality trends. Specifically, we find similar reductions in the counterfactual Gini coefficient when holding the composition fixed with these alternative categorizations.

The similar role of household composition in overall inequality trends implied by both wage-based and hours-based classifications is consistent with their overlap and the role of both career and family objectives in marital sorting. Since we observe less pronounced trends in assortative matching based on hours or lifetime earnings, it is plausible that changes in the composition of households are more pronounced for different-type couples when using the hours-based and lifetime-earnings-based classifications.

For hours, these changes in off-diagonal couple types, in turn, suggest changing patterns of hours specialization. Specifically, increased PAM by ambition happens in two ways that cancel each other out in the aggregate measure for hours: (i) by increasing the share of couples with same ambition and similar hours schedules (e.g., lawyer-lawyer couples) but also (ii) by increasing the share of couples with similar wage dynamics and different schedules (e.g., nurse-carpenter or teacher-nurse couples).

B.4 Counterfactual Scenarios (ii) and (iii) based on Gutierrez (2020)

Table B.2: Scenarios (ii) and (iii) using the Gutierrez (2020) version of Choo and Siow (2006)

	(a) Gini		(b) P_{90}/P_{50}		(c) P_{50}/P_{10}	
Factual change (Δ_{Data})	0.102	100%	0.322	100%	3.626	100%
	Δ_{Gini}	$\frac{\Delta_{Gini}}{\Delta_{Data}}$	$\Delta_{P_{90}/P_{50}}$	$\frac{\Delta_{P_{90}/P_{50}}}{\Delta_{Data}}$	$\Delta_{P_{50}/P_{10}}$	$\frac{\Delta_{P_{50}/P_{10}}}{\Delta_{Data}}$
(ii) Fixed marital surplus						
Educational Level	0.074	73%	0.213	66%	2.263	62%
Educational Field	0.074	73%	0.215	67%	2.207	61%
Educational Ambition	0.069	68%	0.185	58%	2.148	59%
(iii) Fixed marginals						
Educational Level	0.124	122%	0.424	131%	6.020	166%
Educational Field	0.121	119%	0.410	127%	6.011	166%
Educational Ambition	0.102	100%	0.303	94%	4.518	125%
(iii.a) Fixed marginals (male)						
Educational Level	0.101	99%	0.299	93%	4.266	118%
Educational Field	0.100	98%	0.295	92%	4.281	118%
Educational Ambition	0.098	97%	0.298	93%	3.870	107%
(iii.b) Fixed marginals (female)						
Educational Level	0.122	120%	0.392	122%	5.231	144%
Educational Field	0.121	119%	0.392	122%	5.234	144%
Educational Ambition	0.105	103%	0.323	100%	4.201	116%

Notes: Panels show the counterfactual scenarios constructed as explained in Section 5.2 (Panels (ii) to (iii.b)). We modify the Choo and Siow (2006) model following Gutierrez (2020) and do not assume independence of irrelevant alternatives. P_{90}/P_{50} (P_{50}/P_{10}) is the ratio of the 90th and 50th (10th) percentile in the income distribution. Δ_{Data} shows the observed inequality changes. $\Delta_{Gini}/\Delta_{Data}$ is the counterfactual change relative to the observed change. Each row within each scenario shows the columns' statistic for one of the three definitions of types (as explained in Section 3): educational level, educational field, and educational ambition types.

B.5 Details on Robustness

B.5.1 Alternative Construction of Ambition Types

In this section, we assess whether our main results are sensitive to constructing ambition types in alternative ways. Using the notation from the conceptual framework in Section 3.1, recall that our benchmark categorization creates four types using information at the level of the educational program i , the sub-vector of characteristics $\tilde{x}_i = (w_{0i}, g_i)$, and the k-means mapping algorithm $\mathcal{T}_{Ambition}(\tilde{x})$. Table B.3 shows the trends in the degree of assortative matching (analyzed in Section 4) and the inequality contributions in the fixed household composition counterfactual scenario (analyzed in Section 5, scenario (i)) that we find when constructing the ambition types in various alternative ways. For convenience, the first row repeats the results for our benchmark categorization of ambition types. The columns show: (a) the the numbers of observations in 1980 and 2018, (b) the change in our measure of PAM, equation (2), between those years, (c) the observed Gini coefficients in 1980 and 2018, and (d) the counterfactual Gini in the fixed household composition scenario, along with the explained share relative to the trend in the data.

In Panel (i), we analyze the sensitivity of our findings with respect to the assumption that the ambition types do not vary by gender or over time. In the row labeled *Types by gender*, we consider the possibility that education programs may send different signals depending on the gender of the graduate. That is, we construct the four ambition types separately for women and men by using information at the program-gender level i . Formally, we consider the sub-vector of characteristics $\tilde{x}_i^f = (w_{0i}^f, g_i^f)$ for women and $\tilde{x}_i^m = (w_{0i}^m, g_i^m)$ for men and create female and male ambition types through k-means mappings $\mathcal{T}_{Ambition}(\tilde{x}^f)$ and $\mathcal{T}_{Ambition}(\tilde{x}^m)$, respectively. Online Appendix Figure B.6 (which has the same structure as Figure 1) shows that our method successfully generates four types clearly distinct in terms of labor market prospects (as is the case for our benchmark). Even though most of the biggest programs are assigned to the same ambition type for men and women, there are exceptions. For example, architecture is a program associated with a high wage growth type for women but a low wage growth type for men. We find that the degree of assortative matching based on gendered ambition types increased slightly less than in our benchmark (18.6%, which is still significantly more than AM by levels or fields) and that the composition of couples explains slightly less of the changes in inequality than our benchmark does (almost 30%, which is significantly more than what we find using levels or fields). Similarly, for the row labeled *Types by cohort*, we construct ambition types by cohorts of graduates defined by decade (individuals who graduated

Table B.3: Assortative Matching and Fixed Household Composition with Alternative Types

	(a)		(b)			(c)		(d)	
	N (1,000s)		Assortative Matching			Gini, data		Gini, (i)	$\frac{\Delta_{Gini,(i)}}{\Delta_{Gini,data}}$
	1980	2018	1980	2018	Change	1980	2018	2018	
<u>Benchmark types</u>									
Labor income	2,427	2,801	1.29	1.55	20.1%	0.298	0.400	0.364	64.6%
Disposable income	2,427	2,801	1.29	1.55	20.1%	0.227	0.366	0.327	72.3%
<u>(i) Gender- and cohort-specific ambition types</u>									
Types by gender	2,426	2,798	1.15	1.37	18.6%	0.298	0.400	0.374	74.3%
Types by cohort	2,408	2,799	1.34	1.58	18.2%	0.298	0.400	0.372	72.7%
<u>(ii) Different numbers of clusters</u>									
Three ambition types	2,427	2,801	1.23	1.42	15.2%	0.298	0.400	0.373	73.2%
Five ambition types	2,427	2,801	1.39	1.73	24.9%	0.298	0.400	0.361	61.5%
<u>(iii) Different clustering variables</u>									
Only g	2,427	2,801	1.27	1.40	9.9%	0.298	0.400	0.375	75.6%
Only w_0	2,427	2,801	1.35	1.56	15.7%	0.298	0.400	0.375	75.4%
Male w_0, g	1,191	2,775	1.40	1.60	14.2%	0.319	0.401	0.365	56.7%
w_0 , part-time penalty	2,345	2,725	1.37	1.50	9.2%	0.298	0.401	0.369	69.2%
w_0 , manager premium	2,417	2,360	1.46	1.91	31.4%	0.298	0.378	0.354	70.4%
<u>(iv) Different level of aggregation</u>									
Programs $\times$ univ.	2,409	2,795	1.28	1.55	20.7%	0.297	0.400	0.364	65.2%
Sub-fields $\times$ levels	2,427	2,642	1.32	1.51	14.4%	0.298	0.398	0.367	68.5%
<u>(v) Sample Restrictions</u>									
No 3-digit programs	2,403	2,479	1.28	1.50	16.9%	0.298	0.396	0.360	63.4%
Only Ages 40–49	688	885	1.23	1.50	21.9%	0.265	0.311	0.291	56.4%

Notes: Columns (a) display the number of observations in each case in thousands of individuals for both 1980 and 2018. Columns (b) show our measure of assortative matching $\mathcal{S}$ derived in Section 4 for both years along with the percentage change. Columns (c) show the observed Gini coefficients in each case for 1980 and 2018. Columns (d) show the counterfactual Gini coefficient in the fixed marriage market scenario (i), i.e., had the composition of households stayed fixed at its 1980 level in each case and the fraction of the observed change in Gini that is captured by the change in the counterfactual scenario, $\Delta_{Gini,(i)}/\Delta_{Gini,data}$. Each row shows the columns' statistic for one of the alternative definitions of ambition types.

before 1990, between 1990 and 2000, and after 2000). Here, we account for the possibility that the signaling value of degrees may change over time (similar to the Goldin (2014) argument that occupations have evolved over time). We define three sub-vectors of characteristics by graduation cohort, $\tilde{x}_i^{80} = (w_{0i}^{80}, g_i^{80})$, $\tilde{x}_i^{90} = (w_{0i}^{90}, g_i^{90})$, and $\tilde{x}_i^{00} = (w_{0i}^{00}, g_i^{00})$, and map programs to types by cohort using the k-means algorithm. While Online Appendix Figure B.7 shows that many large programs are remarkably stable in their characteristics over time, we detect some changes. For example, while an ordinary high school diploma is categorized as a type with low starting wage and high growth early in the sample, these growth opportunities decline over time and the degree moves into the low-low category. Other changes are based on shifts in relative pay levels, which can also revert back. For example, preschool teachers are classified as low-low in the first and last part of the sample, but fall into the high starting wage and low

growth category in the 1990s. Both our main conclusions regarding the changes in assortative matching and the relationship between household composition and inequality are unchanged when constructing the ambition types by cohort.

In Panel (ii), we analyze the sensitivity of our findings with respect to the assumption that there are exactly four ambition types. To this end, we repeat the analysis but define three and five ambition types instead of four. Once again, our main conclusions remain the same. While more (fewer) categories lead us to detect a slightly stronger (weaker) increase in assortative matching on educational ambition of 24.9% (15.2%), our conclusions on the role of household composition for rising inequality remain.

In Panel (iii), we vary the clustering variables that we use to construct ambition types. In the first two rows, we only use one of our benchmark clustering variables, either wage growth g (the sub-vector of characteristics becomes $\tilde{x}_i = (g_i)$) or starting wages w_0 ($\tilde{x}_i = (w_{0i})$), respectively. In the third row, we use both clustering variables, but use only male outcomes to construct ambition types for both men and women. The concern behind this robustness check is that the endogenous labor supply choices of females, e.g., in response to child birth (Kleven et al., 2019) could bias our way of capturing the signaling value of educational programs by affecting the average starting wages and, especially, wage growth of graduates. Therefore, we only use the starting wages and wage growth of male workers, who have more stable labor supply trajectories, to cluster programs. Finally, in the fourth row, we replace wage growth with one of our direct measures of career flexibility and work-life balance based: the part-time penalty (a measure of inflexibility, recall Section A.5). In all four robustness checks, we find that ambition types constructed with alternative clustering variables continue to reveal significant increments in terms of assortative matching (9.2%–15.7%) and that changes in households' education composition explain significant shares of increasing between-household inequality (33.6%–52.6%). What all these alternative categorizations have in common is that they use a sub-vector of characteristics that correlates with both the earnings potential and the work-life balance attached to the unit of observation, as shown in Table A.3. Thus, our conclusion that AM on career ambition has increased over time and that changes in the marriage market contribute significantly to the rise in between-household income inequality is robust to constructing ambition types differently.

In Panel (iv), we assess the robustness of our findings to constructing ambition types at different levels of aggregation. First, the row labeled *Programs* $\times$ *university* uses a level of observation that is more granular than the education programs used for our benchmark categorization. Specifically, we distinguish between similar programs offered at different Danish uni-

versities and define $i' = \text{programs} \times \text{university}$ as our unit of observation for the categorization. As our clustering variables, we use the same sub-vector as in our benchmark, $\tilde{x}_{i'} = (w_{0,i'}, g_{i'})$. The assortative matching trend and the explained share of the inequality increment by household composition in this more disaggregated version of our approach are almost identical to the benchmark. This result is consistent with Appendix Figure B.8, which shows that labor market outcomes w_0 and g are very similar across graduates of the same degree program across universities. Second, the row labeled *Sub-fields* $\times$ *levels* uses a level of observation that is more aggregate than the granular education programs used for our benchmark categorization. We aggregate programs by levels and sub-fields of study and define $i' = \text{levels} \times \text{sub-fields}$ as our unit of observation. Specifically, we consider 48 observation units that we obtain by subdividing each of the four educational levels by sub-field of study. Essentially, sub-fields are a more detailed version of the fields of study used above.²⁴ The results only change slightly. Assortative matching increased by about 14% and the household composition continues to explain around 40% of the increasing inequality between households.

B.5.2 Additional Sample Restrictions

In Panel (v) of Table A.3 we provide robustness of the main results to additional sample restrictions. First, we consider a reduced sample that omits all individuals with coarse educational program information. Coarse information is defined as observing only 3-digit degree information rather than 4-digit educational degrees. The latter is standard for Danish citizens but we observe a significant share of immigrants with only 3-digit degree information. Hence this robustness test is informative about the role of immigrants for our results. This specification yields quantitatively similar results for inequality to the baseline: The household composition accounts for 36.6% of the increase in inequality, and the PAM trend is slightly muted, with an increase in assortative matching by 16.9%. Intuitively, part of the increase could be driven by marriages among immigrants who are, based on their coarse educational information, more likely assigned to the low ambition cluster. The second specification in Panel (v) focuses only on individuals aged 40-49, following the cohort approach used in the literature (see e.g., Chiapori et al. (2020a)). While this sample is less representative, the evidence shows that our main findings on assortative matching and inequality trends are robust and that any changes in the composition of young individuals is not driving our main results.

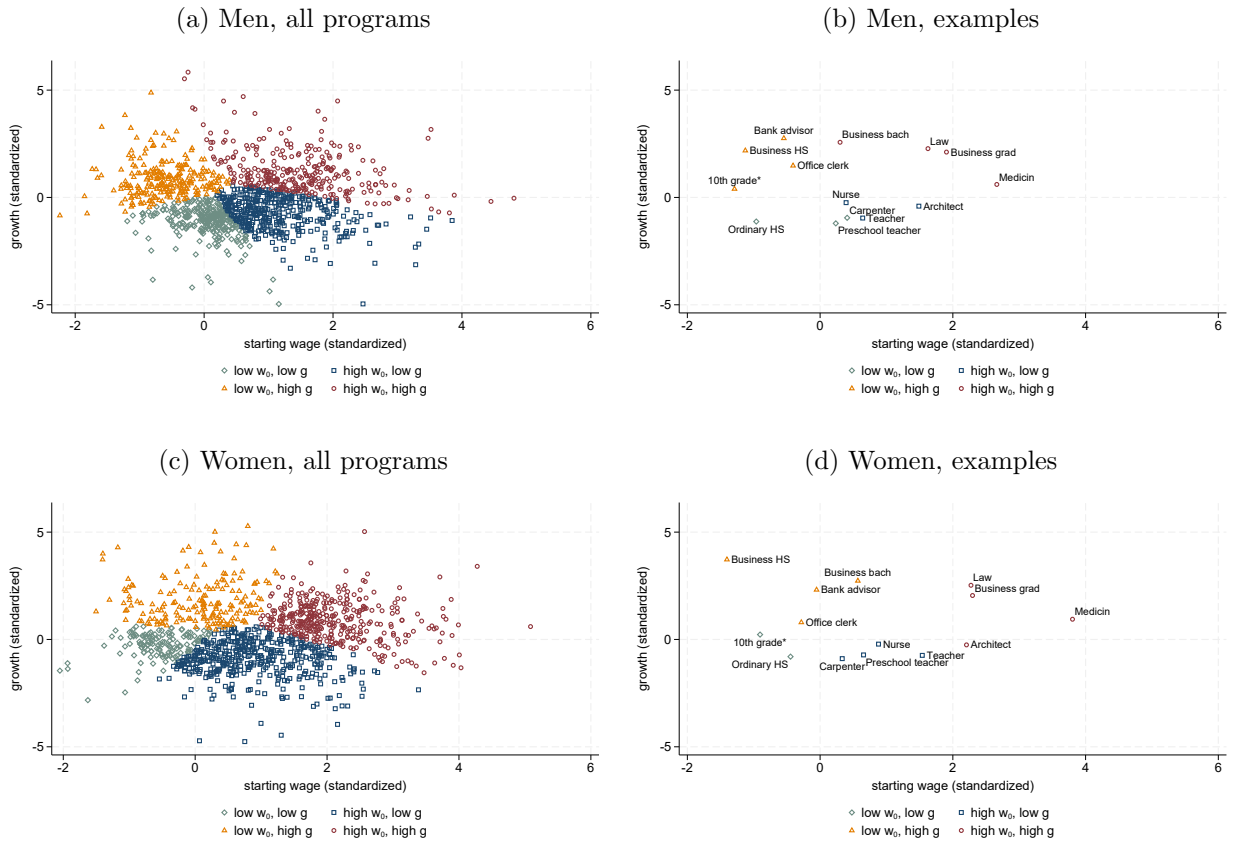
²⁴For example, we consider the STEM sub-fields “Construction” and “Mechanics & Metal” and further distinguish programs within this sub-field by the required level of schooling for entry into the programs, i.e., high school (secondary programs) and college/university (tertiary programs).

B.5.3 Disposable Income

Finally, Table A.3 provides a comparison of our baseline results for labor income to inequality trends when using disposable income instead. Crucially, this post-tax measure also includes government transfers. We again apply the OECD equivalence scale to analyze inequality in total disposable income across households. The results of this analysis are reported in the top panel for easy comparison with the baseline. Despite a lower level of inequality, we find a substantial increase in disposable income inequality in the data. 27.7% of this increase can be accounted for by the household composition based on our benchmark ambition types. This evidence shows that redistribution through taxes and transfers does not substantially mitigate the role of marital sorting in explaining inequality trends.

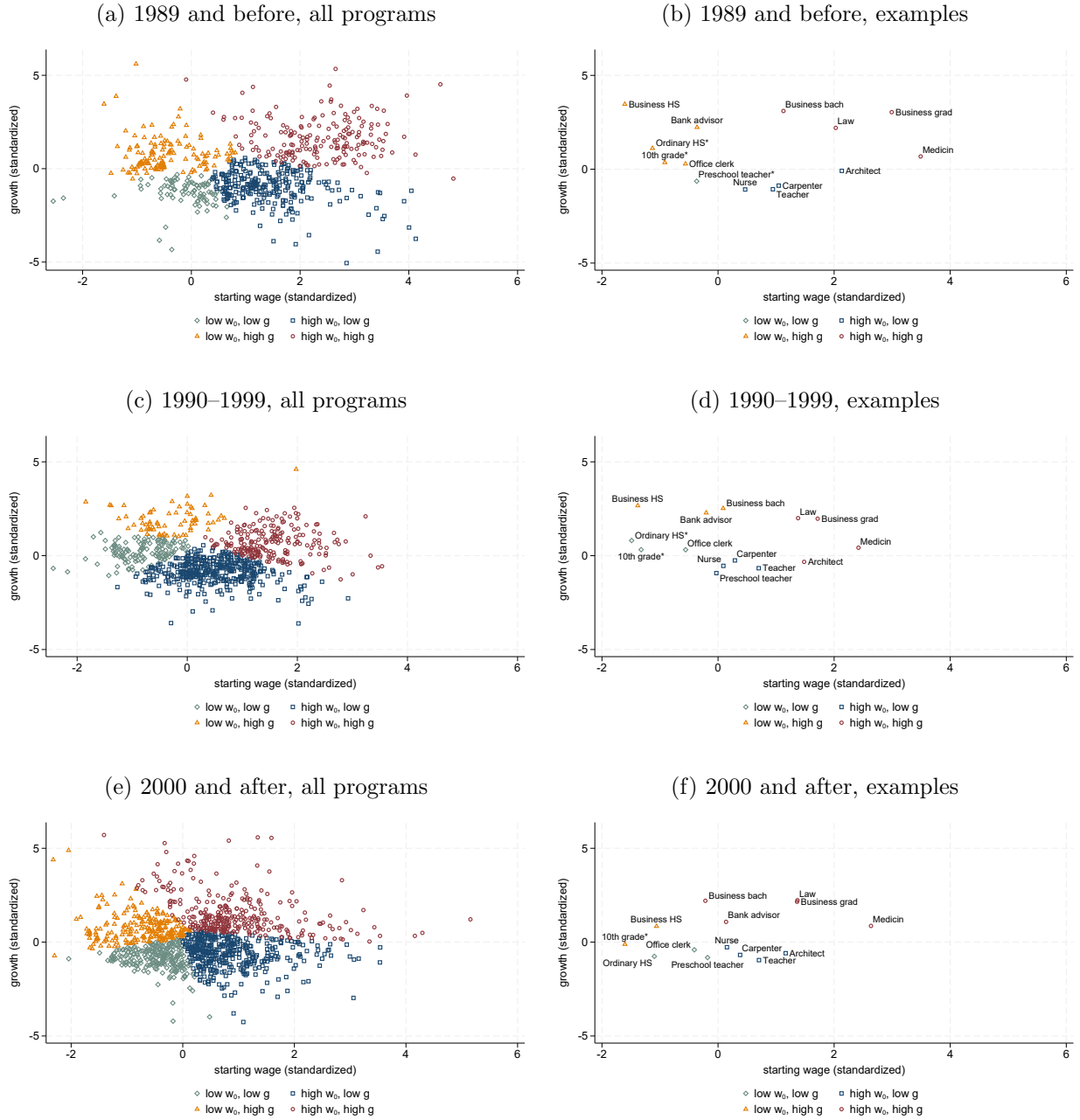
B.5.4 Supporting figures for robustness analysis

Figure B.6: Educational Program Categorizations by Gender



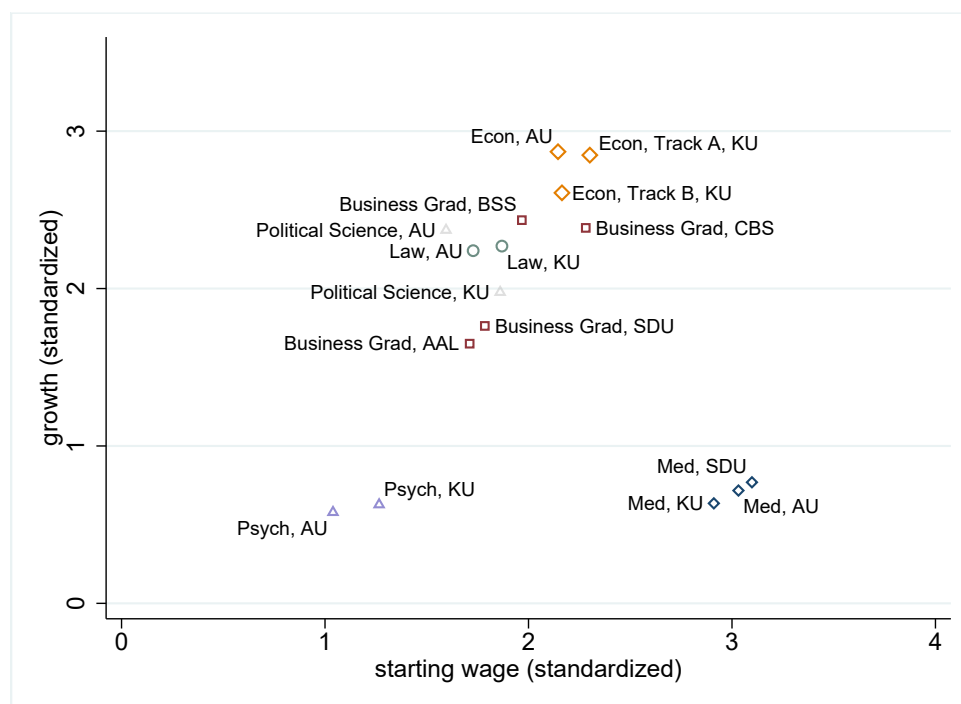
Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The horizontal axis corresponds to the standardized w_0 and the vertical to the standardized g . Points in the panels locate all educational programs with at least ten graduates in 2018—described in Section 2—along these two dimensions. Colors and markers uniquely assign each program to a marriage market type, depending on the panel's definition.

Figure B.7: Educational Program Categorizations by Decade



Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The horizontal axis corresponds to the standardized w_0 and the vertical to the standardized g . Points in the panels locate all educational programs with at least ten graduates in 2018—described in Section 2—along these two dimensions. Colors and markers uniquely assign each program to a marriage market type, depending on the panel's definition.

Figure B.8: Starting Wages and Wage Growth for Main Degree Programs by University



Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The horizontal axis corresponds to the standardized w_0 and the vertical to the standardized g . Points in the panels locate degree programs by university that have at least 1000 graduates 1980–2018 along these two dimensions. Same symbols and colors indicate the same degree program across different universities.

Online Appendix References

- Almar, Frederik and Bastian Schulz (2024) “Optimal weights for marital sorting measures,” *Economics Letters*, Vol. 234, p. 111497.
- Calvo, Paula, Ilse Lindenlaub, and Ana Reynoso (2024) “Marriage Market and Labour Market Sorting,” *The Review of Economic Studies*, Vol. 91, pp. 3316–3361.
- Chiappori, Pierre-André, Monica Costa-Dias, Sam Crossman, and Costas Meghir (2020) “Changes in Assortative Matching and Inequality in Income: Evidence for the UK,” *Fiscal Studies*, Vol. 41, pp. 39–63.
- Choo, Eugene and Aloysius Siow (2006) “Who marries whom and why,” *Journal of Political Economy*, Vol. 114, pp. 175–201.
- Goldin, Claudia (2014) “A Grand Gender Convergence: Its Last Chapter,” *American Economic Review*, Vol. 104, pp. 1091–1119.
- Gutierrez, Federico H (2020) “A simple solution to the problem of independence of irrelevant alternatives in Choo and Siow marriage market model,” *Economics Letters*, Vol. 186, p. 108785.
- Kleven, Henrik, Camille Landais, and Jakob Egholt Søgaaard (2019) “Children and Gender Inequality: Evidence from Denmark,” *American Economic Journal: Applied Economics*, Vol. 11, p. 181–209.
- Lund, Christian Giødesen and Rune Vejlin (2016) “Documenting and Improving the Hourly Wage Measure in the Danish IDA Database,” *Nationaløkonomisk tidsskrift*, Vol. 2016, pp. 1–35.
- Wiswall, Matthew and Basit Zafar (2021) “Human Capital Investments and Expectations about Career and Family,” *Journal of Political Economy*, Vol. 129, pp. 1361–1424.